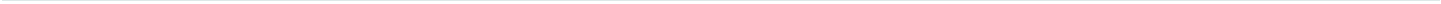




The whole system, end to end.

Riziki Industrial Chemicals sells chemicals by the kilo, mixes them into products, gives credit to laundries and car washes, and needs to know at the end of the day whether any of it made money. This is how the system does that, in the order it happens.

One database A single file on one server. Nothing leaves the country.	Two roles Owner and attendant, plus six permissions granted by name.	Works offline The counter keeps selling when the network drops.	Nothing is deleted Mistakes are corrected by further entries, in the open.
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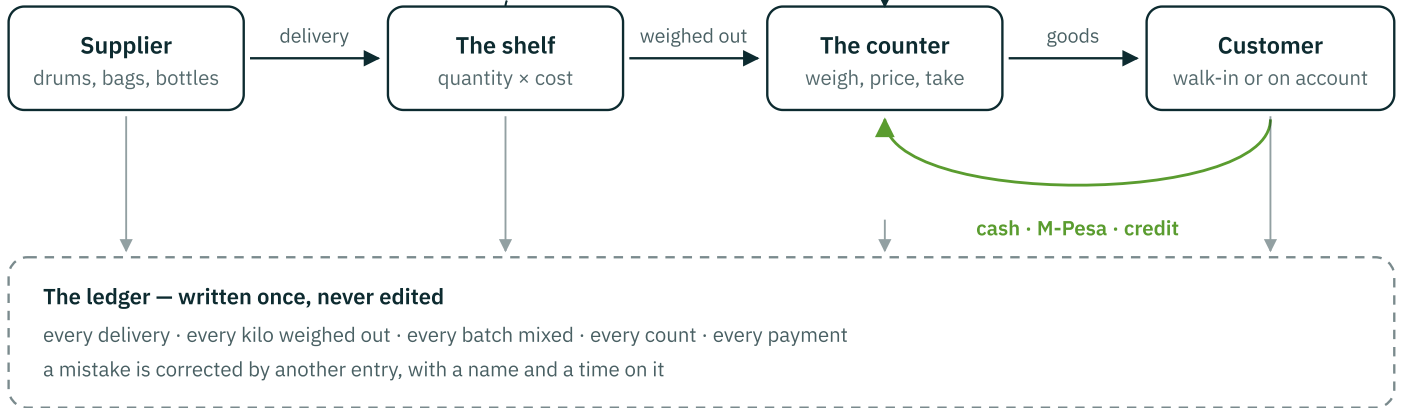
A • THE SHAPE OF IT

Goods come in one side, money goes out the other, and a ledger runs under both.

Three things move through the shop and each is recorded separately: the **goods**, which arrive from a supplier and leave over the counter; the **money**, which arrives when the customer pays — not necessarily the same day; and the **ledger**, which writes down every movement of both and is never edited afterwards.

Almost every question the owner asks is one of those three read back a different way.

or mixed into a product first



Goods run left to right. Money comes back in green, and may come back later than the goods left — which is why the reports count sales on the day the goods went and money on the day it arrived. Everything drops into the ledger underneath.

B · A DAY AT THE COUNTER

Sign in, sell, take the money, close the day.

The counter is a phone on a stand with a queue in front of it. Everything about that screen is built around the queue: no confirmation dialogs, no page reloads, no waiting for a network.

- 1 **Sign in with a PIN.** Each person has their own. Every sale, price change and void carries the name of whoever did it.
- 2 **Find the item.** Tap it on the colour-coded board, or type its street name — "sles", "labsa", "20 caustic ⚡". Recipes appear beside the chemicals.
- 3 **Say how much.** Kilos, litres or pieces. The price per unit is the price the shop quotes, so nothing has to be worked out in anybody's head.
- 4 **Haggle, within limits.** The price can come down to the floor the owner set. Below it, the screen asks for the owner's PIN — and the discount is recorded either way.
- 5 **Take the money.** Cash, M-Pesa, or part of each on one bill. An M-Pesa code can only be used once, so one SMS can never pay two bills.
- 6 **Hand over the paper.** Thermal receipt over Bluetooth, an A5 sheet, or the invoice straight into WhatsApp as a file.
- 7 **Close the day.** Count the drawer. The system says what it expected, and the difference is recorded rather than argued about.



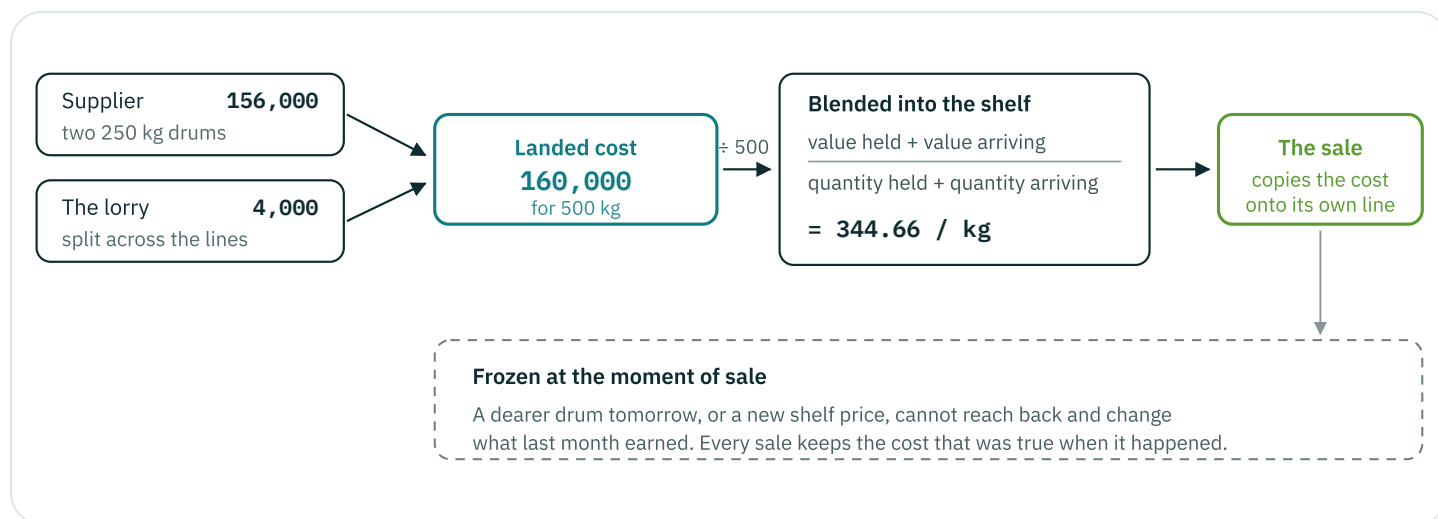
The counter on the shop phone. Products the shop mixes sit above the chemicals it sells neat. Figures shown are sample data.

When the network drops, the counter keeps selling. Sales are held on the phone and sent up when the line returns, each one carrying an id that makes sending it twice harmless.

C • THE GOODS PATH

Cost arrives with the goods, and never gets typed in by hand.

When a delivery is entered, the system does three things at once: it puts the quantity on the shelf, works out what that quantity cost including its share of the transport, and blends that into the cost already on file. From then on, every kilo on that shelf carries a price the shop can defend.



The costing chain, with the real figures from the worked example below. Cost is always per kilo, litre or piece — never per drum, because the same chemical arrives in 250 kg drums and in 200 kg drums, and the average cost of a drum is the cost of nothing.

The shelf can also be changed without a customer

Three operations move stock without a sale, and each writes its own entry: **repacking** a drum into jerricans, with the spillage recorded rather than hidden; **mixing** a batch, which takes chemicals off and puts a product on; and a **stock take**, where what is counted becomes the new truth and the difference is reported as shrinkage.

D • RECIPES AND MIXING

A recipe can be sold two ways, and the books handle both.

Every formula is versioned and scales to any batch size — ask for 200 litres and the quantities come out in kilos and litres, with the mixing steps beside them. What differs is when the mixing happens.

ROUTE ONE — MIXED IN ADVANCE

A batch is mixed on the mixing board and lands on the shelf as real, counted stock. The chemicals leave and the product arrives in one movement, so nothing is counted twice. The cost of everything that went in divides by **what the drum actually held** — not by what the recipe hoped for.

BATCH OF MULTIPURPOSE	QUANTITY	AT COST	VALUE
Ungerol	10 kg	344.66	3,446.60
Ufacid	5 kg	350.02	1,750.10
Salt	10 kg	59.94	599.40
Caustic Soda, C.M.A, colour	0.6 kg	—	449.63
Perfume	10 pcs	150.22	1,502.20
What went in			7,747.93
What came out — the drum held 196 L, not 200	196 L		39.53 / L

Twenty litres then sells at KES 220 a litre: 4,400.00 charged, 790.60 of cost, **KES 3,609.40 of profit**. The mix is on the shelf like any other product, and can run low, be counted, or go dead.

ROUTE TWO — MIXED TO ORDER

Nothing is mixed until somebody asks. The counter sells a size, and the chemicals come off the shelf at that moment. The sale is written as one priced line for the product and unpriced ingredient lines underneath it — the receipt shows one charge, the stock shows seven chemicals leaving, and neither is double counted.

WHAT THE SALE WRITES DOWN	CHARGED	COST
Multipurpose — 20 L bundle	4,400.00	0.00
· Ungerol	0.00	344.66
· Ufacid	0.00	175.01
· Salt	0.00	59.94
· Caustic Soda, C.M.A, colour	0.00	44.95
· Perfume	0.00	150.22
Together	4,400.00	774.78

KES 3,625.22 of profit — the reports add the two halves together. Reading those rows as they lie would say the mix was free to make and the chemicals were given away, which is exactly the bug that made mixed products show a 100% margin. It is fixed, and a test now fails if the halves ever come apart again.

Online

Mixing board

Carwash ShampooFabric SoftenerMultipurpose Cleaner

Carwash Shampoo

one batch makes 20 L - sells at KES 300/L

HOW MUCH DID YOU MAKE, IN L

The recipe is written for 20 L. Set the sizes it is sold in on Carwash Shampoo, under Products & prices, to count it by the jerrican instead.

Say how much Carwash Shampoo the batch made.

Adjust what went in or came out

Record this batch

Tap a size to say how much Carwash Shampoo you made.

ON THE SHELF147 L

MIXED RECENTLY

WHEN	BATCH	MADE	COST OF WHAT WENT IN
27 Sept 2026, 16:59	B-260927-3 Fabric Softener - Owner Undo this batch	117.6 L Fabric Softener from 3 kg G. Base, 6 kg Flares, 0.12 kg Colour (assorted), 6 pcs Perfume	KES 4,566
27 Sept 2026, 16:59	B-260927-2 Carwash Shampoo - Owner Undo this batch	147 L Carwash Shampoo from 7.5 kg Unipend, 7.5 kg Salt, 0.375 kg Simet, 0.188 kg C.D.E	KES 3,547.76
27 Sept 2026, 16:59	B-260927-1 Multipurpose Cleaner - Owner Undo this batch	196 L Multipurpose Cleaner from 10 kg Unipend, 5 kg Unisid, 10 kg Salt, 0.2 kg Caustic Soda, 0.2 kg C.M.A, 0.2 kg Colour (assorted), 10 pcs Perfume	KES 8,100

The mixing board. Below the form, every batch already run, what it made, and what the chemicals in it cost. Sample data.

Why the yield is typed in

The shop dilutes by eye with a hosepipe. A system that insists a 200-litre recipe made exactly 200 litres is a system that gets lied to. Record 196 and the four litres that never appeared are carried by the ones that did — which is why the same 20 litres costs 38.74 a litre made to order and 39.53 from a batch.

E • CREDIT AND THE WHOLESALE SIDE

Money arrives later than the goods, and the books say which day is which.

A laundry takes forty litres on Tuesday and pays on the following Monday. Both facts matter and they belong to different days: the **sale** is Tuesday's, the **money** is Monday's. The dashboard shows them separately rather than averaging them into one misleading figure.

A quote

Priced up, printed or sent as a PDF, and approved by the customer before anything moves.

An invoice

What the quote becomes. Carries the customer's terms, and what has been paid against it so far.

A delivery note

Goes on the lorry with the goods. Signature block, and a line saying the goods stay the shop's until they are paid for.

A statement

Every bill and payment for one customer, with the balance. Ready to send on WhatsApp.

Each customer has a credit limit, and the debt ages by the day. The oldest unpaid bill is on the first screen the owner opens — not buried three taps down, because a debt nobody looks at is a debt nobody collects.



Riziki Industrial Chemicals

Nairobi
0722 000 000

DELIVER TO

Eastleigh Cleaners Ltd
0722 418 330

DELIVERY NOTE
DN INV-00002

19 Sept 2026, 19:43

SALE

INV-00002
Served by Owner

Address / destination

Vehicle / driver

#	ITEM	QUANTITY
1	20 L jerrican	1 pcs
2	Acid Thickener	1 (18.5 kg)
3	B.G	23.5 L

3 lines · this note is for the goods only. The invoice carries the money.

Received the above goods in good order and condition.

RECEIVED BY (NAME)

SIGNATURE

DATE

DELIVERED BY

Goods remain the property of Riziki Industrial Chemicals until paid for in full. Any shortage or damage must be reported on the day of delivery.

A delivery note as it prints. Sample data.

Seven rules, and a day you can add up yourself.

- 1 Cost is per kilo, litre or piece.** Never per drum.
- 2 Cost arrives with the goods.** Supplier price plus that line's share of the transport.
- 3 Each delivery blends into the shelf.** A running weighted average, always current.
- 4 The cost freezes onto the sale.** Nothing later rewrites what a past month earned.
- 5 An unknown cost is reported as unknown.** Not as free goods. The screen says how much it had to estimate.
- 6 A batch puts all its money onto what came out.** Water carries mass but not cost.
- 7 Gross = sales – cost of goods. Net = gross – expenses.** Margin is gross over sales.

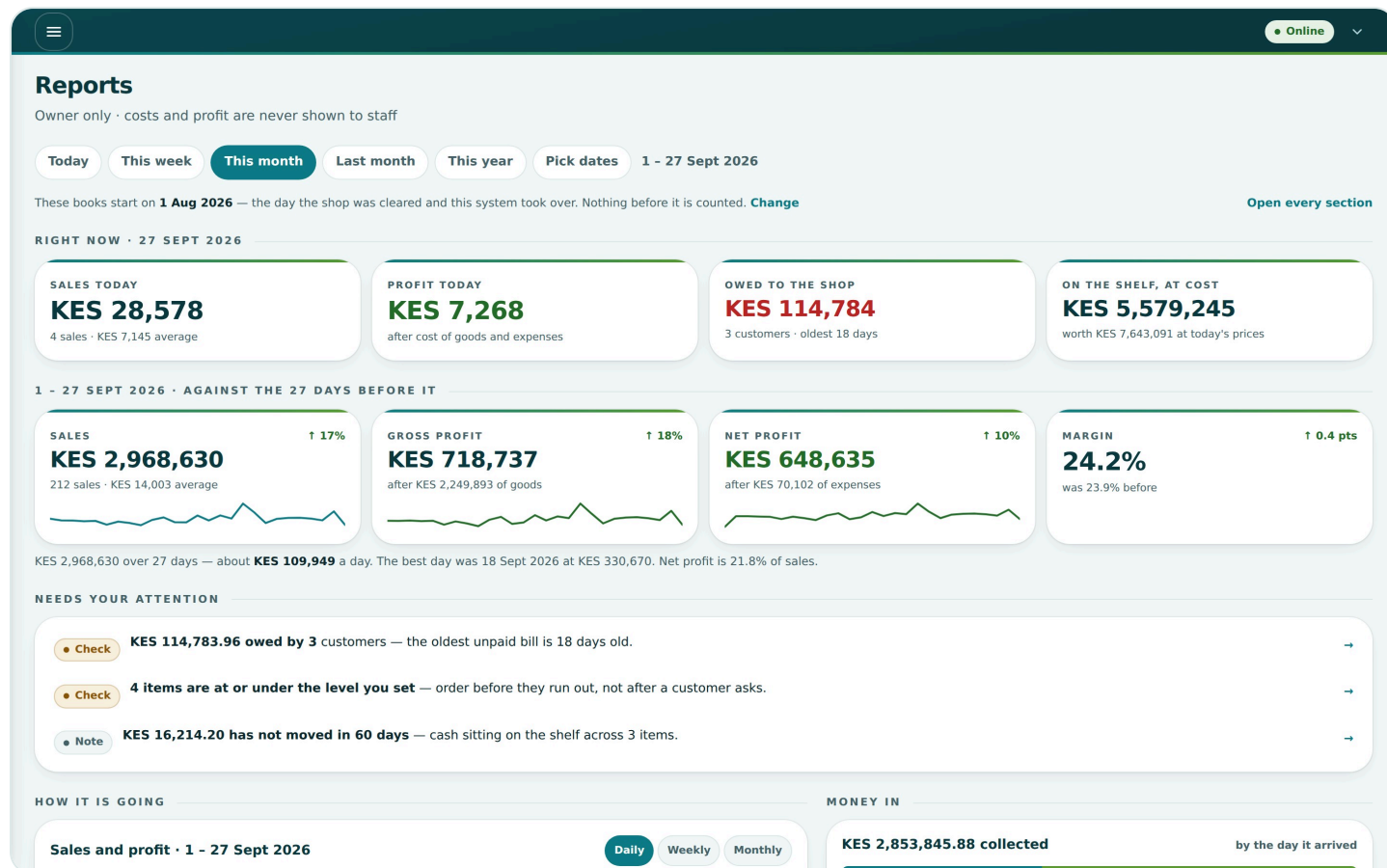
ONE DAY, WORKED THROUGH	SOLD	COST	PROFIT	MARGIN
Ungerol, 20 kg over the counter	10,260.00	6,400.00	3,860.00	38%
Multipurpose, 20 L mixed to order	4,400.00	774.78	3,625.22	82%
Multipurpose, 20 L from the batch	4,400.00	790.60	3,609.40	82%
Gross profit	19,060.00	7,965.38	11,094.62	58%
Expenses — one delivery run		1,200.00		
Net profit			9,894.62	52%

Every figure above is printed by a script that buys, sells and mixes through the same code the shop runs, so the documentation cannot drift away from the system. The margins are high because these are the sample recipe's prices, not the shop's — the arithmetic is the point.

The rows must always add up to the total. If the per-product list ever misses the period's own figure by a cent, something is being counted twice or lost — so the test suite checks that equality on every build rather than trusting it.

Ninety seconds, standing up.

Not a report to read — a check to make, in the order an owner makes it. Everything below the first screen folds away and remembers what was left open.



The dashboard as it opens. Sample data.

Right now

Today's takings, today's profit, what customers owe, and what the shelf is worth — true at this second, whatever period is selected.

Against the days before

Every figure carries the same number of days immediately before it, and the shape of the period drawn underneath.

Needs your attention

Worked out, not written: debts and their age, stock running low, anything selling below cost. Each one links to the screen that fixes it.

How it is going

Sales as bars, profit as a line, on one plot. Daily, weekly or monthly — selling more and keeping more are two different questions.

Money in

Counted by the day it arrived: cash against M-Pesa, credit given, old bills settled. Deliberately not the same figure as sales.

What earns

Per product and per line of business, with the losing rows showing what they sold at and what they are costed at, per kilo.

Under those: the day-by-day ledger, the whole book since the day the shop was cleared, discounts by person and by item, dead stock, shrinkage — and exports to PDF, a spreadsheet per table, or the whole database.

H • WHO SEES WHAT

Two roles, and six switches on top.

An attendant sells. An owner sees everything. That covers a shop of two people and stops covering it the moment there is a third — a manager who records deliveries and counts stock but has no business knowing what a drum cost. So permissions are granted to one person at a time, by name.

PERMISSION	WHAT IT OPENS	WHO HAS IT
Cost prices and profit	Reports, margins, what the shop paid.	GRANTED
Recipes and the mixing board	What goes into each mix, and mixing a batch.	GRANTED
Products and prices	A shelf price, a floor, a ceiling, a size. Not the cost.	GRANTED
Record deliveries	What arrived from a supplier, and what it cost.	GRANTED
Count the stock	The count sheet, and where a count came from.	GRANTED
Void a sale	Cancel one rung up wrong. Both entries are kept.	GRANTED
Any of the six, on an owner	An owner passes every check by being one. Ticking boxes on an owner does nothing.	ALWAYS
Selling, and yesterday's receipts	The counter, the sales list, a customer's balance.	EVERYBODY
Accounts, the activity log, exports, backup	How an owner watches everything above.	OWNER ONLY, NEVER GRANTED

Every check runs on the server before the data is read. A staff phone does not receive cost figures and then hide them — it never receives them. A permission that only hides a menu item is not a permission.

I • WHAT CANNOT BE CHANGED

The four rules that make the

numbers worth believing.

The ledger is append-only

The database itself refuses an edit or a delete on stock movements, sales, price changes and pack counts. A correction is another entry, with a name and a time on it. Nobody — including whoever runs the server — quietly rewrites history.

A sale is voided, never deleted

The stock comes back, the money comes back, and both entries stay on the record with the reason written down.

Prices and costs are snapshots

Each sale line keeps the price asked, the price charged and the cost, as they were that day. Re-pricing tomorrow cannot change what last month earned.

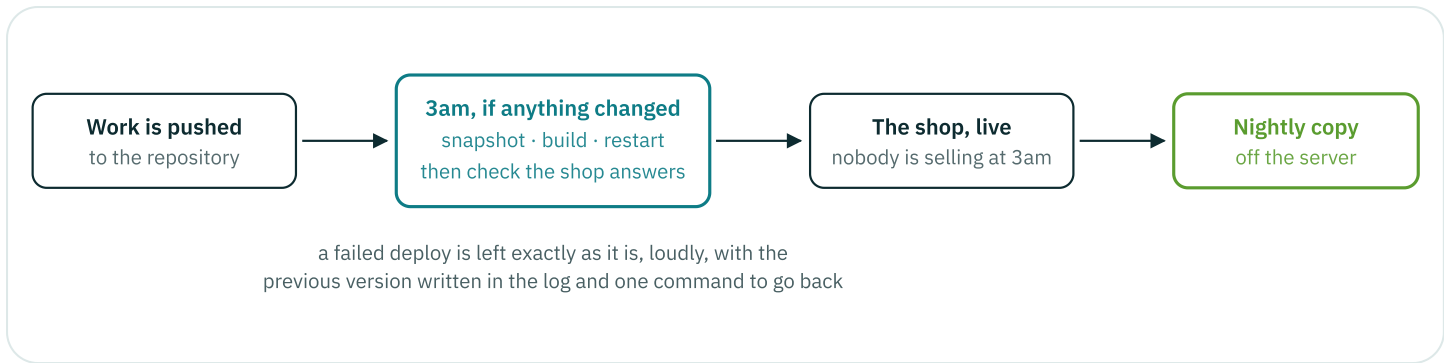
The books have a start date

The shop was trialled, emptied and started again. Every report counts from the day it was cleared, and says so on the screen.

J • KEEPING IT RUNNING

One server, one file, and a copy of it somewhere else every night.

The whole shop — every sale, every chemical, every recipe — is one database file small enough to email. That is deliberate: a shop that cannot carry its own records is a shop renting them.



Deployment and backup. The overnight job does nothing at all when there is nothing to deploy, and says so in the log.

If the shop is slow or down

One health check script says which part is not answering. Restarting is one command; going back to the previous version is one more.

If a figure looks wrong

The activity log says who did what and when. Every report exports to a spreadsheet, so anything can be checked outside the system.

If the server is lost

Restore last night's copy onto a new one. The runbook has the commands, written for somebody who has never seen the system.

If nobody technical is available

The runbook is written to be followed by the owner. Nothing in it needs the person who built it.

Riziki POS · built for Riziki Industrial Chemicals, Nairobi. Every screen on this page is the running system photographed against a demonstration shop — the trading figures are invented, and the shop's real numbers are not in this document.